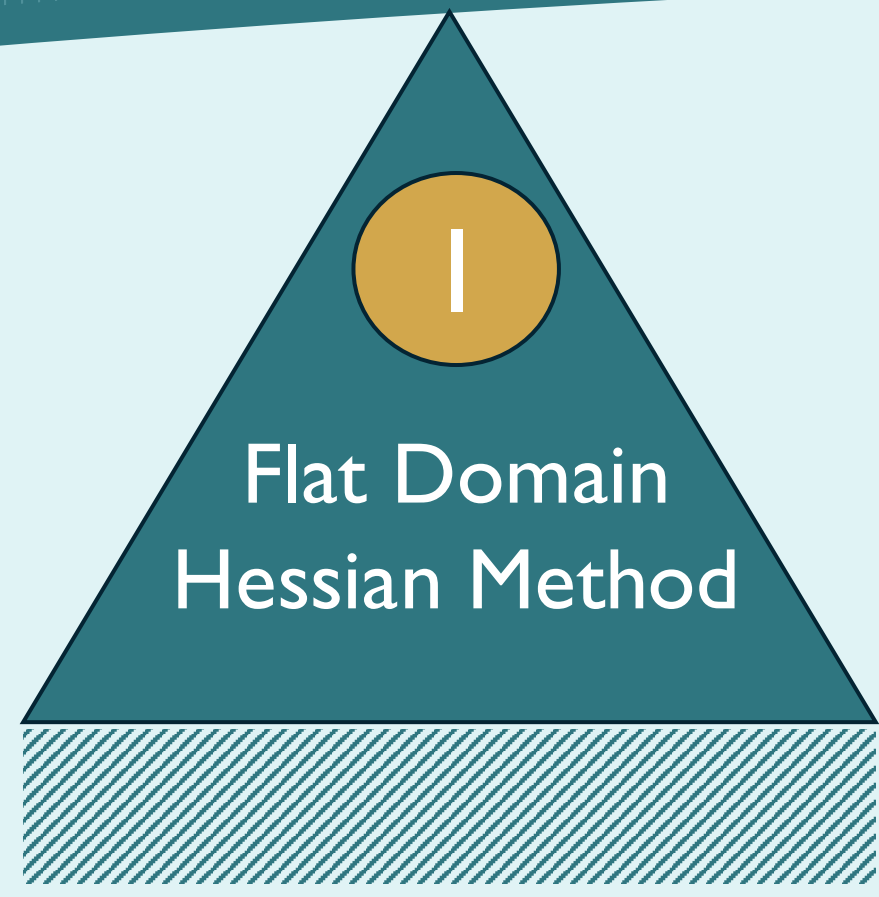


A Numerical Comparison of Two Discrete Models of Surface Bending

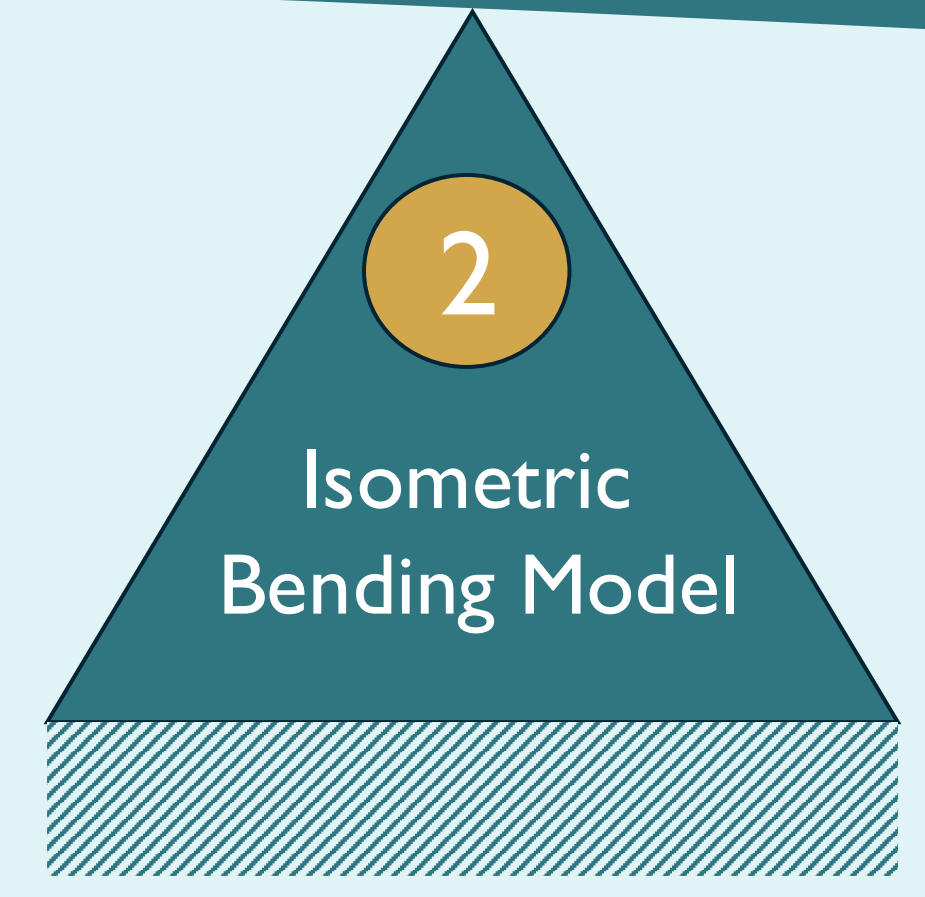
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0 Introduction

Thin surfaces bend all around us, from sheets of paper to biological membranes enclosing cells. Understanding this bending has applications ranging from cloth dynamics to simulations of exosomes. How can we model the forces that resist it? This resistance can be described by a change in curvature energy, whose variation gives the bending force density. In simulations, however, surfaces are represented by piecewise-flat triangular meshes, where curvature is not directly defined. This project compares two discrete models of surface bending by examining their computation times and convergence under mesh refinement.



1 Geometry of an Embedding

Our story begins with embedding a surface S in $\mathbb{R}^3$. This embedding takes the form $\mathbf{X}(u^1, u^2) \in \mathbb{R}^3$

The metric to the surface is

$$g_{ij} = \langle \mathbf{e}_i, \mathbf{e}_j \rangle$$

The second fundamental form and shape operator are

$$b_{ij} = -\mathbf{n} \cdot \partial_i \partial_j \mathbf{X}, \quad S_j^i = g^{ik} b_{kj}.$$

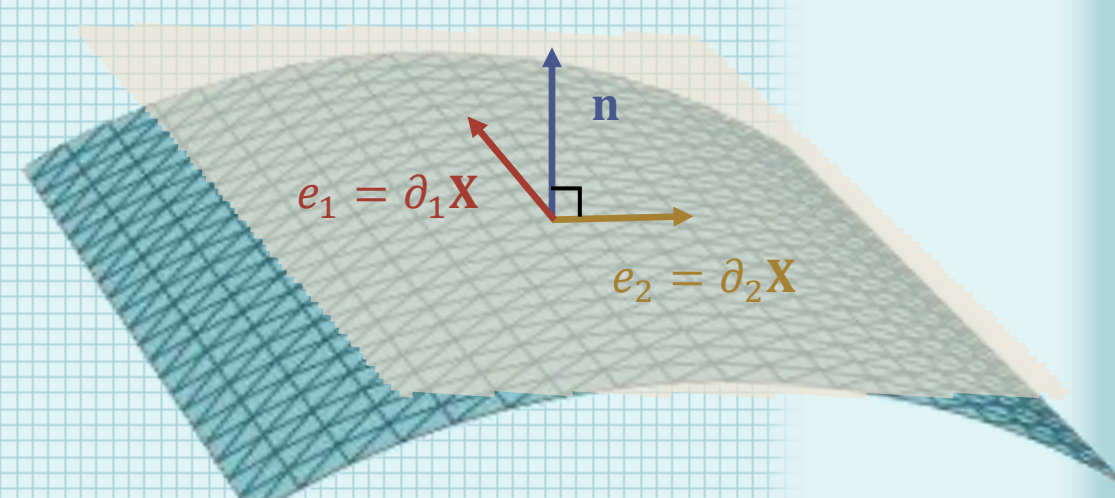
The eigenvalues of (S_j^i) are the principal curvatures k_i and the mean curvature is defined as

$$H = \frac{(k_1 + k_2)}{2}$$

An important relation between the Laplace-Beltrami operator surface and H is

$$\Delta_S \mathbf{X} = -2H \mathbf{n},$$

where $|g| = \det(g_{ij})$.



2 From Curvature to Force

Curvature is represented by the tensor b_{ij} , whose invariants are the *trace* and the *determinant*. These two invariants are the mean curvature H and the Gaussian curvature K .

$$\text{tr}(b_{ij}) = k_1 + k_2 = 2H, \quad \det(b_{ij}) = k_1 k_2 = K$$

Squaring and integrating the mean curvature H gives an orientation-independent bending energy that penalizes changes in curvature, where κ is the bending modulus. This energy functional is the primary study of this project

$$E = \frac{\kappa}{2} \int_S (2H)^2 dA$$

Using the Laplace-Beltrami identity, the energy can be written as

$$E[\mathbf{X}] = \frac{\kappa}{2} \int_S \langle \Delta_S \mathbf{X}, \Delta_S \mathbf{X} \rangle^2 dA$$

The variational derivative of this energy functional gives the sought after bending force density

$$\mathbf{f}_b = \frac{\delta E}{\delta \mathbf{X}}$$

3 Two Discrete Bending Models

1 Flat Domain Hessian Energy (FDHE)

3.1

FDHE

- Discrete smoothness energy based on the Hessian of a height field.
- Models bending energy for small deformations of a flat domain
- Energy:

$$E_H = \int_S \|\nabla^2 h\|^2 dA$$

- Discrete form:

$$E_h = \frac{\kappa}{2} h^T Q h \Rightarrow \mathbf{F}_H = -\kappa Q h$$

where $Q = B^T M^{-1} B$

- Limitations: only applicable to a flat domain with small deformations

2 Isometric Bending Model (IBM)

3.2

IBM

- Discrete bending model for isometric deformations
- It discretizes the Laplace-Beltrami operator
- Energy:

$$E[\mathbf{X}] = \frac{\kappa}{2} \int_S \langle \Delta_0 \mathbf{X}, \Delta_0 \mathbf{X} \rangle^2 dA_0$$

- Discrete form:

$$E_b = \frac{\kappa_b}{2} \mathbf{X}^T Q_b \mathbf{X} \Rightarrow \mathbf{F}_b = -\kappa_b Q_b \mathbf{X}$$

where $Q_b = L^T M^{-1} L$

- The mean curvature vector is

$$2H \mathbf{n} \approx M^{-1} L \mathbf{X}$$

- Limitations: only applicable for isometric deformations

4 Test Case

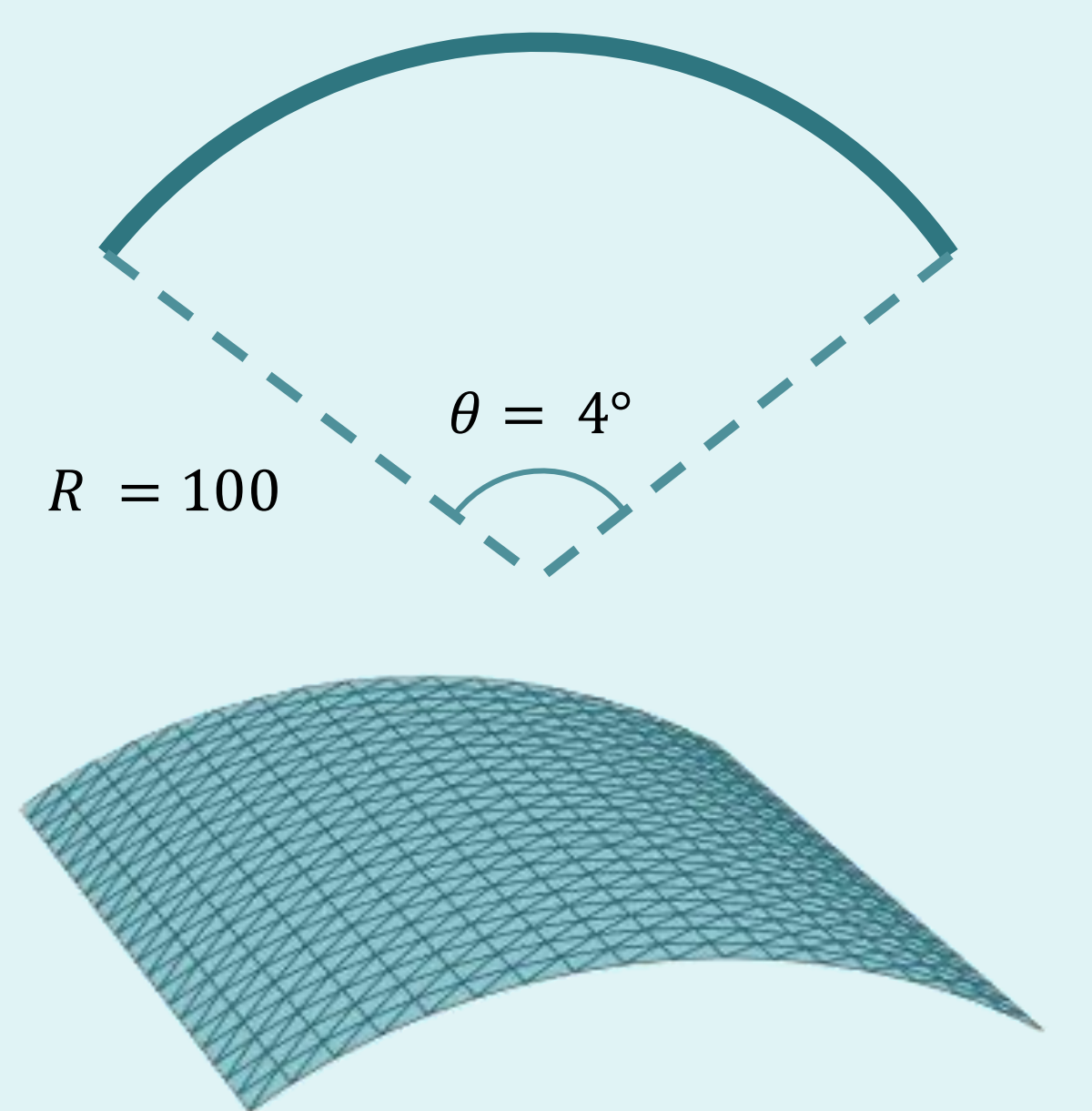
We study a shallow arc with radius $R = 100$, straight axis length $L = 100$, and angular length $\theta = 4^\circ$.

Its principal curvatures are

$$k_1 = 1/R, \quad k_2 = 0$$

Giving $H = 1/2R = 1/200$ (and $K = 0$)

The continuum force densities were derived as the variational derivatives of the respective energies for this test case. These provide the analytical values against which the numerical force densities are compared.



6 Results

- Arc test: mean absolute error across all mesh resolutions remained below 10^{-12} for curvature: **relative error below $2 \cdot 10^{-8}\%$**
- Error remained below $3.5 \cdot 10^{-10}$ for normal force density: **relative error below 0.035%**
- The icosphere test results indicate near **second order convergence** in average vertex area
- Both methods show **linear time complexity** in the number of vertices N .
- The IBM constructs Q faster (by a factor of around 3)

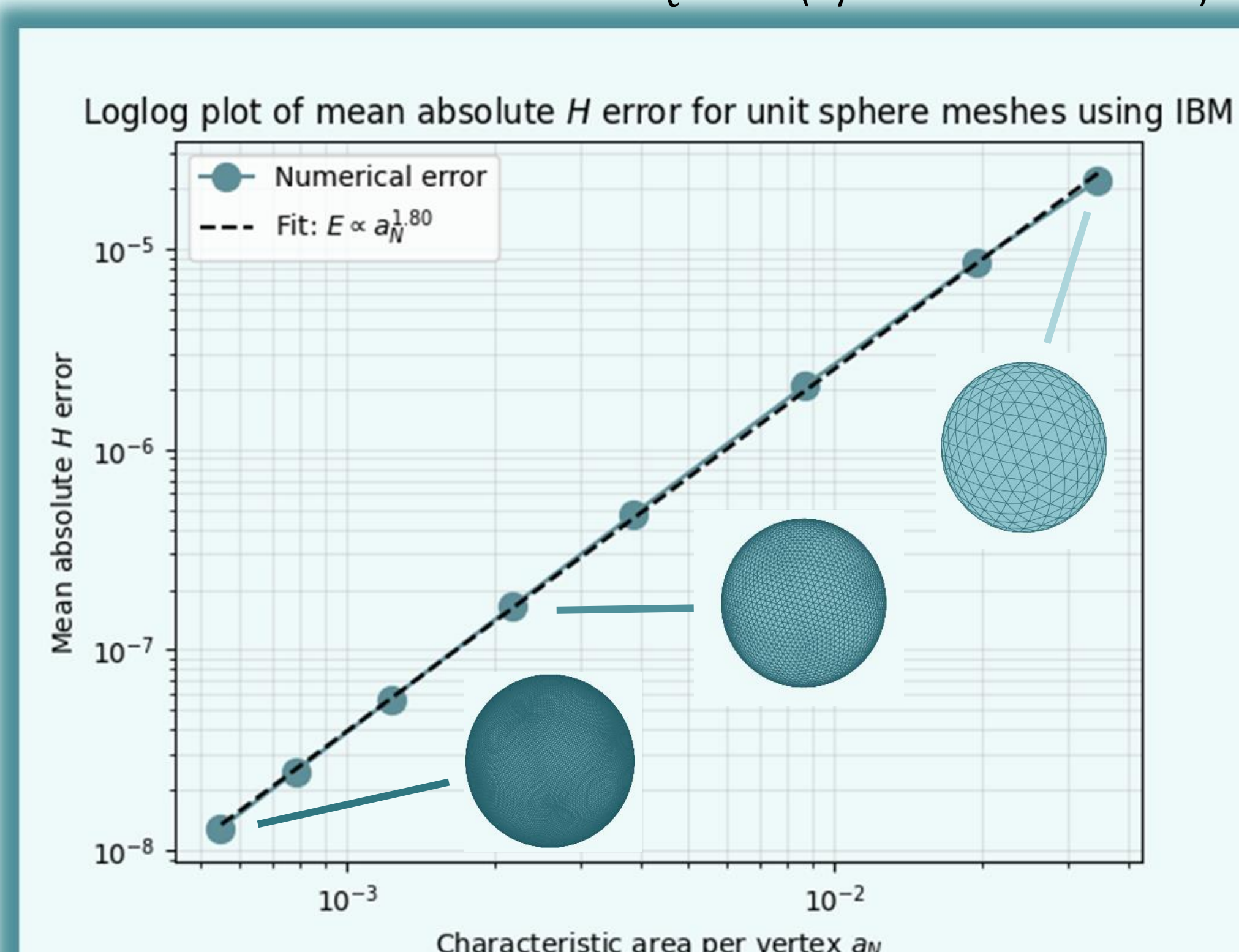
5 Method

- The arc was discretized using a uniform grid at eight mesh resolutions. The discrete models supply nodal forces which are converted into force densities using the nodal areas.
- Since the analytical expressions describe the interior Euler-Lagrange force, errors were evaluated only on vertices sufficiently far from the patch boundary.
- Mean-curvature convergence was tested separately on refined unit-sphere meshes,
- Construction times for Q using both methods were averaged over ten trials.

7 Conclusion & Remarks

- Both models closely reproduced the analytical bending-force densities with low relative errors. The Laplacian-based mean curvature estimate converged under refinement, and construction time for the discrete operators scaled linearly for both models.

- The results suggest that both models can accurately represent bending in their intended settings, with the IBM being **particularly attractive for cloth dynamics**, where stretching is negligible.
- An application closer to home, in our own bodies, is the modelling of fluid membranes. Extending the IBM to these surfaces requires accounting for how the discrete operators change as the membrane moves. This opens an exciting direction: **using geometry to understand how microscopic membranes bend, flow and reshape within our bodies.**



# of vertices	FDHE (ms)	IBM (ms)
438	24.824	8.497
1908	102.102	31.878
7920	430.369	136.379
32352	1857.6	556.647